HW #5

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1. Reading-Related Question

(a) The primitive soup hypothesis (Oparin-Haldane) says life began in a dilute ocean where UV light and lightning produced organic compounds like amino acids and nucleotides. A key weakness is specificity: the soup has no way to ensure that polymers form with the correct chemical bonds. Proteins require peptide bonds and RNA requires specific nucleotide linkages, but Miller-type experiments produce a mix of bonds. Wrong linkages block replication entirely.

The primitive pizza hypothesis (Wächtershäuser) says life began on positively charged iron pyrite surfaces. Organic ions bond to the surface and are held in a fixed orientation, able to move only in two dimensions instead of freely in solution. This increases both reaction speed and specificity, and concentrates molecules so they can react. This directly solves the soup's problem because the surface enforces correct orientations for polymerization.

(b) Without replicase enzymes the error rate is very high, too high to replicate any meaningful length of molecule reliably (the error threshold). With replicase but without proofreading, the error rate is 1/1000 to 1/10,000 per nucleotide. This is because RNA strands separate immediately after copying, so there is no way to check whether each base was paired correctly. Replicase improves accuracy significantly, but without proofreading errors are still too frequent for large genomes, which is expected since enzyme geometry alone improves selectivity but cannot catch all mistakes.

(c) RNA is single-stranded, so there is no complementary strand to use for proofreading. With an error rate of 1/1000 to 1/10,000 and a human genome of 3 billion bases, every replication cycle would introduce millions of mutations. RNA is also chemically unstable since its $2'$-OH group makes it susceptible to hydrolysis. A human with RNA genes would rapidly accumulate errors that destroy genetic information faster than it can be replicated, making stable complex life impossible.

2. ^{12}C–^{13}C Fractionation

(a)

$$m_{\text{proton}} = m_{\text{neutron}} = 1.67 \times 10^{-27} kg$$
$$m_{^{12}\text{C}} = 12 \times 1.67 \times 10^{-27} \text{ kg} = 2.004 \times 10^{-26} \text{ kg}$$
$$m_{^{13}\text{C}} = 13 \times 1.67 \times 10^{-27} \text{ kg} = 2.171 \times 10^{-26} \text{ kg}$$

(b) For a particle in a box of width $L = 0.2$ nm $= 2 \times 10^{-10}$ m:

$$E_n = \frac{n^2 h^2}{8mL^2}$$

For $n = 1$, with $h^2 = (6.626 \times 10^{-34})^2 = 4.390 \times 10^{-67}$ J^{2}s^2 and $L^2 = (2 \times 10^{-10})^2 = 4 \times 10^{-20}$ m^2:

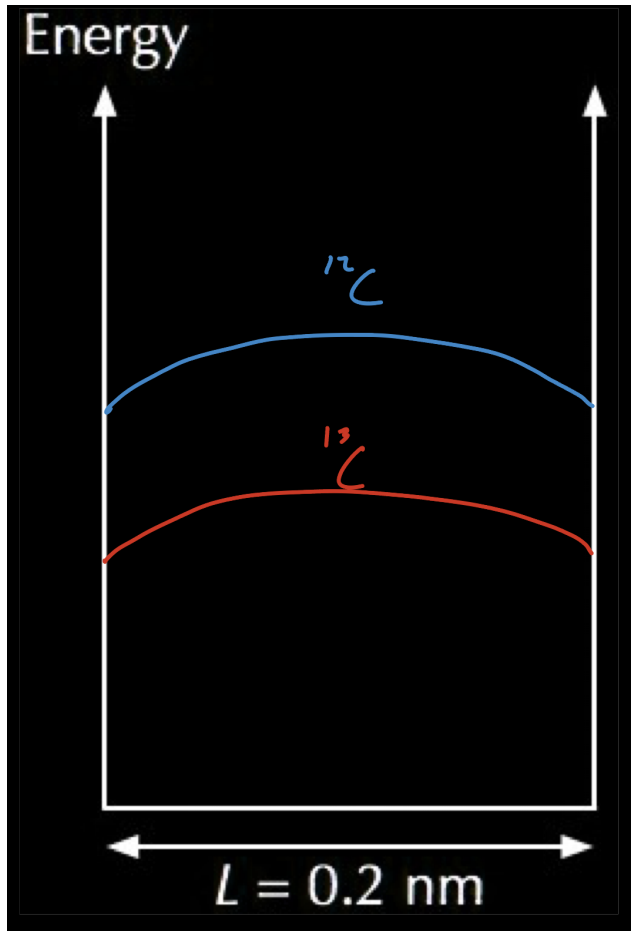

Figure 1: TODO: add caption

$$8\, m_{^{12}\mathrm{C}}\, L^2 = 8 \times 2.004 \times 10^{-26} \times 4 \times 10^{-20} = 6.413 \times 10^{-45}$$

$$E_1(^{12}\mathrm{C}) = \frac{4.390 \times 10^{-67}}{6.413 \times 10^{-45}} = 6.85 \times 10^{-23}\ \mathrm{J} \approx 0.427\ \mathrm{meV}$$

$$8\, m_{^{13}\mathrm{C}}\, L^2 = 8 \times 2.171 \times 10^{-26} \times 4 \times 10^{-20} = 6.947 \times 10^{-45}$$

$$E_1(^{13}\mathrm{C}) = \frac{4.390 \times 10^{-67}}{6.947 \times 10^{-45}} = 6.32 \times 10^{-23}\ \mathrm{J} \approx 0.394\ \mathrm{meV}$$

The $n = 1$ wavefunction is the lowest energy state with no internal nodes. Both isotopes have the same wave shape, but because ^{12}C is lighter, it has a higher ground-state energy.

Note: The $n = 1$ energy is called the zero-point energy (or ground state energy). It is the minimum energy a quantum particle can have. Unlike a classical particle, which could sit still at zero energy, quantum mechanics requires the particle to always be moving due to the uncertainty principle, so there is always some minimum energy even in the lowest state.

(c) Let E_{ts} be the (same) transition-state energy for both isotopes. The activation energy for each is:

$$E_a = E_{\mathrm{ts}} - E_1$$

Since rate $\propto e^{-E_a/k_BT}$, the ratio is:

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = \frac{e^{-(E_{\text{ts}}-E_1^{12})/k_BT}}{e^{-(E_{\text{ts}}-E_1^{13})/k_BT}} = e^{(E_1^{12}-E_1^{13})/k_BT}$$

$$\Delta E = E_1(^{12}\text{C}) - E_1(^{13}\text{C}) = 6.85 \times 10^{-23} - 6.32 \times 10^{-23} = 5.3 \times 10^{-24} \text{ J}$$

$$k_BT = (1.381 \times 10^{-23})(300) = 4.14 \times 10^{-21} \text{ J}$$

$$\frac{\Delta E}{k_BT} = \frac{5.3 \times 10^{-24}}{4.14 \times 10^{-21}} = 1.28 \times 10^{-3}$$

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{1.28\times10^{-3}} \approx \boxed{1.00128}$$

^{12}C reacts about 0.13% faster than ^{13}C, consistent with observed isotope fractionation in photosynthesis.

(d) At higher temperature k_BT increases, so $\Delta E/k_BT$ decreases and the rate ratio approaches 1. The fractionation effect weakens and both isotopes react at nearly the same rate.

(e) For ^{13}C to react faster, the transition state must also have zero-point energy, with $E_{\text{ts}}(^{12}\text{C}) > E_{\text{ts}}(^{13}\text{C})$ by more than the ground-state difference. This happens when the bond to carbon is stiffer in the transition state than in the reactant, giving ^{12}C a larger energy penalty at the transition state so its activation energy ends up higher.

3. First Cells

(a) Replication inside the vesicle consumes free nucleotides and converts them into RNA polymers that cannot leave. This drops the internal nucleotide concentration below the outside concentration, so nucleotides diffuse in down the gradient.

(b) Water follows osmosis from low solute concentration to high. More RNA polymers inside means lower water concentration inside, so water flows in until osmotic pressure balances membrane tension.

(c) Vesicles with more RNA have higher osmotic pressure so they pull in more water and fatty acids, growing at the expense of low-RNA vesicles which shrink. This is natural selection at the vesicle level: better replicators grow and take resources from worse ones, linking RNA replication efficiency to vesicle survival.

(d) Without helicase or any other machinery, strand separation could be achieved by thermal cycling. In a hydrothermal environment the temperature cycles between hot and cold. At high temperatures the hydrogen bonds between strands break and the strands melt apart. When it cools, each strand can serve as a template for a new copy.

4. Handedness

(a) Using the Boltzmann distribution with $\Delta E = 10^{-17}$ eV (L favored):

$$\frac{P(L)}{P(D)} = e^{\Delta E/k_BT}$$

$$k_BT = (8.617 \times 10^{-5} \text{ eV/K})(295 \text{ K}) = 0.02542 \text{ eV}$$

$$\frac{\Delta E}{k_B T} = \frac{10^{-17}}{0.02542} = 3.93 \times 10^{-16}$$

$$\frac{P(L)}{P(D)} = e^{3.93\times10^{-16}} \approx 1 + 3.93 \times 10^{-16} \approx \boxed{1.000\,000\,000\,000\,000\,4}$$

The L-form is favored by only about 1 part in 10^{16}, so the two forms are essentially equally probable at room temperature.

(b) We want $P(D)/P(L) = 0.1$, so $P(L)/P(D) = 10$:

$$e^{\Delta E/k_B T} = 10 \qquad \Rightarrow \qquad k_B T = \frac{\Delta E}{\ln 10} = \frac{10^{-17}\ \text{eV}}{2.303} = 4.34 \times 10^{-18}\ \text{eV}$$

$$T = \frac{4.34 \times 10^{-18}\ \text{eV}}{8.617 \times 10^{-5}\ \text{eV/K}} \approx \boxed{5.04 \times 10^{-14}\ \text{K}}$$

This is far below any temperature achievable in nature.

(c) No. The effect at room temperature is only $\sim 10^{-16}$, which is nowhere near enough to explain the near-100% L-amino acid homochirality seen in biology.

(d) Another explanation is amplification through autocatalysis: a small random chiral imbalance gets amplified because one enantiomer catalyzes its own production. Circularly polarized UV light from neutron stars could also preferentially destroy one form, creating an initial bias that then gets amplified.